# Rajat Kumar

*Software Development Engineer (SDE) | Full-Stack Developer | MERN Stack*

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## Professional Summary

Results-driven **Software Development Engineer (SDE)** and **MCA graduate** with full-time production experience building and shipping scalable web applications at a **funded product-based company**. Specialized in **frontend development** using **React.js, JavaScript (ES6+), TypeScript,** and **Next.js**, with hands-on expertise in **responsive web design**, **component architecture**, **performance optimization**, and **cross-browser compatibility**. Proficient across the **MERN stack** (MongoDB, Express.js, React.js, Node.js) with strong skills in **RESTful API development**, **state management** (Redux Toolkit), **Git/GitHub workflows**, and **Agile/Scrum methodologies**. Proven track record of resolving production defects, delivering pixel-perfect UI implementations, and contributing to end-to-end **software development lifecycle (SDLC)**.

## Technical Skills

**Languages:** JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL

**Frontend:** React.js, Next.js, Redux Toolkit, Tailwind CSS, Responsive Web Design, DOM Manipulation, Webpack

**Backend:** Node.js, Express.js, RESTful API Development, Mongoose ODM, JWT Authentication

**Databases:** MongoDB, SQL (MySQL), Redis

**DevOps & Tools:** Git, GitHub, Docker, Postman, Vercel, CI/CD Pipelines, Linux CLI

**Cloud & Services:** Cloudinary (Media APIs), Razorpay (Payment Gateway), Nodemailer

**Methodologies:** Agile/Scrum, Software Development Life Cycle (SDLC), Test-Driven Development (TDD), Code Review, Cross-Browser Testing, Manual & Regression Testing

**Architecture & Concepts:** Component Architecture, State Management, MVC Pattern, REST Architecture, Role-Based Access Control (RBAC), System Design

## Professional Experience

**Funded Firm — Product-Based CRM Platform** 6 Months
*Software Development Engineer (SDE) — Full-Time* *Full-Time*

- Engineered, optimized, and shipped **responsive UI components** for a production-grade **CRM platform** using **React.js** and **Tailwind CSS**, resolving **30+ layout, alignment, and viewport defects** across mobile and desktop breakpoints.
- Developed **modular, reusable frontend components** and screens from Figma design specifications, adhering to the established **component architecture** and **design system** guidelines, reducing UI development time by ~20%.
- Identified, reproduced, and resolved **production frontend defects** and visual inconsistencies across **Chrome, Firefox, Safari, and Edge**, improving cross-browser **application reliability** and user experience.
- Executed comprehensive **manual testing** and **regression testing** across core user flows, systematically verifying edge cases, navigation paths, and form validations prior to production releases — reducing post-release bugs by ~35%.
- Conducted and participated in **peer code reviews** via **Git/GitHub pull request workflows**, enforcing clean code standards, modularity, and architectural best practices across the frontend codebase.
- Built and integrated **backend API endpoints** using **Node.js** and **Express.js**, validating request/response payloads, status codes, and error handling using **Postman** for end-to-end API testing.
- Collaborated with cross-functional engineering teams in an **Agile/Scrum** environment, participating in sprint planning, daily standups, and retrospectives to analyze feature requirements and deliver on sprint goals.
- Optimized frontend **performance** through lazy loading, code splitting, and efficient **state management** with React hooks, resulting in measurably faster page load times.

## Projects

**StudyNotion — Full-Stack Ed-Tech Platform** | *React.js, Redux Toolkit, Node.js, Express.js, MongoDB, Tailwind CSS, Razorpay, Cloudinary, JWT* Live Demo

- Architected and deployed an end-to-end **full-stack educational platform** with **Role-Based Access Control (RBAC)** supporting Students, Instructors, and Administrators — serving dynamic content to 3 distinct user roles.
- Implemented secure **user authentication and authorization** using **JSON Web Tokens (JWT)**, bcrypt password hashing, and OTP-based email verification via **Nodemailer**, ensuring industry-standard security practices.
- Integrated **Razorpay payment gateway** for seamless course purchases, automated enrollment workflows, and secure transaction processing with webhook verification.
- Developed a comprehensive **Admin Dashboard** featuring course catalog CRUD controls, user management (with cascading deletion), and **revenue analytics** visualization using **Chart.js**.
- Designed and implemented **RESTful CRUD APIs** in **Express.js** with **MongoDB/Mongoose**, integrated **Cloudinary SDK** for cloud-based media asset storage and streaming, and deployed the application live on **Vercel**.
- Managed global application **state** using **Redux Toolkit** with async thunks, implementing optimistic updates and centralized error handling for a seamless user experience.

## Education

**Master of Computer Applications (MCA)** 2024 – 2026
*Vinoba Bhave University, Hazaribagh | CGPA: 7.8*

**Bachelor of Computer Applications (BCA)** 2021 – 2024
*Annada College, Hazaribagh | CGPA: 6.41*

**12th (Senior Secondary) — Science** 2020
*IGSI College, Mandu*

**10th (Secondary School)** 2018
*Chumba High School*

## Core Competencies

Problem Solving & Analytical Thinking | Agile/Scrum Methodologies | Cross-Browser Testing & QA | Responsive UI/UX Development | Performance Optimization | Clean Code & Code Review | RESTful API Design & Integration | State Management | Version Control (Git/GitHub) | Technical Documentation